Dreaming LoRA

A Biologically-Motivated Memory Architecture for Transformers

Ben Sovocool
Developed in collaboration with Claude (Anthropic)

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Abstract

We propose a memory architecture for transformer-based language models that bridges short-term and long-term memory through a biologically-motivated consolidation mechanism analogous to sleep. The architecture uses the existing K/V cache as high-resolution, transient working memory; iteratively-updated LoRA adapters as persistent long-term memory encoded as structural deformations of attention geometry; and a "dreaming" consolidation process that transfers information between the two via prediction-error-weighted, noise-regularized gradient extraction. The consolidation objective is formalized as dimensionality reduction of the systematic gradient signal across generative dream variations, implemented via truncated SVD on per-dream gradient matrices with respect to the query and value projection weights. The per-dream loss is composite: a reconstruction term (next-token prediction) and a representational coherence term (contrastive alignment across dream contexts), with the balance self-regulated by the spectral energy of the mean gradient modulated by adapter maturity. Homeostatic weight decay is applied directly to the LoRA factors after the SVD update, enabling rejuvenation without competing with experiential signal for the rank-constrained gradient subspace. Dream content is generated through *adapter-biased context transplantation*: salient K/V content is embedded in alien contexts selected via importance sampling over the adapter's receptive field, with an anti-bias correction that upweights contexts the adapter has learned to avoid. We analyze the stability properties of the resulting co-constitutive dynamics—in which the adapter shapes attention, which shapes future prediction errors, which shape future adapter updates—using the framework of constant-stepsize stochastic approximation with Markovian noise, and argue that the appropriate stability target is bounded tracking (ergodic concentration around a stable set) rather than fixed-point convergence. Five architectural components are identified, each with a distinct functional role and a biological correlate: the K/V cache (working memory), the LoRA adapter (cortical consolidation), the dreaming process (sleep replay), a norm-bounded trust region (trauma prevention), and a shadow adapter via Polyak–Ruppert averaging (stable identity/character). The proposal synthesizes and extends recent work on test-time memory (Titans), generative replay (Dream2Learn), sleep-like consolidation (SRC), and continual learning with LoRA, while contributing a novel integration: structured dreaming as directed search over gradient-space perturbations, with dream content functioning as an importance-sampling distribution biased by prediction error magnitude.

1 Introduction: The Memory Problem

Current transformer-based language models have no mechanism for experiential memory that persists across conversations and modifies processing. The dominant approaches to memory—context injection (RAG, memory summaries) and parametric fine-tuning—each fail in characteristic ways.

Context injection inserts propositional memory into the input sequence. This makes memory accessible but not operative: the injected propositions are processed through the same undifferentiated attention mechanism as all other tokens. They do not reshape how the model attends to new information, do not create predictive expectations, and do not function as the kind of internalized narrative that shapes perception in biological cognition. Memory-as-context is memory as a reference text consulted during processing, not memory as a structural feature of the processing system itself.

Parametric fine-tuning (including LoRA) can produce persistent modifications to model behavior, but as currently deployed it occurs offline, is driven by external training objectives rather than the model's own experience, and lacks any mechanism for gradual, stability-preserving consolidation. Naive fine-tuning on new data produces catastrophic forgetting—the destruction of previously acquired knowledge by overwriting the weight structures that encoded it [9].

The biological solution is well-characterized at a functional level: a layered memory system with distinct short-term and long-term components, connected by a gradual consolidation process (sleep-dependent replay) that transfers information from fast, high-resolution episodic encoding to slow, distributed, structural modification of processing dynamics [13]. This paper proposes an analogous architecture for transformers, grounded in the computational logic of biological memory consolidation but specified in terms of existing transformer components. The technical core of the proposal is a consolidation objective defined as low-rank approximation of the systematic gradient signal across dream variations, driven by a composite loss whose reconstruction/coherence balance is self-regulated (§3), a concrete dream-generation mechanism specified as adapter-biased context transplantation (§5), and stability properties analyzed via constant-stepsize stochastic approximation (§4). The self-reinforcing dynamics of the consolidation loop require that the adapter have existing structure before self-directed dreaming becomes productive, which motivates a three-phase developmental trajectory from externally-supervised curriculum through transitional exploration to mature self-directed consolidation (§5.7).

2 Architecture Overview

The architecture comprises five components, each with a distinct functional role.

2.1 K/V Cache as Short-Term Memory

The transformer's key-value cache accumulates within a conversation, storing individually preserved past-state snapshots that flow horizontally across token positions and are selectively retrieved via attention ($Q \cdot K$ similarity). It is high-resolution, temporally local, and transient—discarded when the conversation ends. Functionally, it is the analog of hippocampal fast encoding: a vivid, complete, but impermanent record of recent experience.

2.2 LoRA Adapter as Long-Term Memory

A LoRA adapter [8] applies a low-rank perturbation ($\Delta W = BA$, where rank r is small) to the query (Q) and value (V) projection matrices of the attention mechanism.

This modifies how the model queries the K/V cache (via W_Q) and what information it extracts (via W_V). Crucially, this does not store propositions. It deforms the metric of the attention comparison space—changing what the model treats as similar to what, and what it retrieves from working memory. This is memory as altered perception, not memory as stored content.

The adapter persists across conversations and constitutes the structural residue of accumulated experience. All updates are applied to the LoRA adapter while the base model remains frozen, making the adapter a learned correction field on top of a fixed prior.

2.3 Dreaming as Consolidation

The missing piece in current architectures is a mechanism connecting short-term to long-term memory. We propose a consolidation process, executed between conversations, that transfers information from the K/V cache into the LoRA adapter through structured generative replay—"dreaming."

After a conversation, the system generates noisy variations of the high-salience content from the K/V cache (identified by within-conversation prediction error; see §3). These variations are produced through a concrete generative mechanism—adapter-biased context transplantation—specified in §5. The variations are processed through the current model-plus-adapter state, producing per-variation loss signals and corresponding gradient matrices with respect to W_Q and W_V. The stable structure across these gradient matrices—what survives averaging over variations—is extracted and applied as the adapter update.

The noise is not random. It is structured, generated by the model itself, conditioned on the high-salience regions of the K/V cache, and biased toward exploring the regions of perturbation space where the current topology is most uncertain or most stressed. Dream content is the search function: the character of the imagined variations navigates gradient space via importance sampling biased by prediction error magnitude. This is computationally analogous to the emotional biasing of biological dream content, where emotionally salient experiences are preferentially replayed and recombined during sleep [4].

2.4 Norm-Bounded Trust Region

Each consolidation step is bounded not only in rank (directional complexity) but in operator norm (step intensity). This prevents any single experience, regardless of how surprising, from producing a deformation larger than the surrounding adapter structure can absorb. This is the architectural analog of trauma prevention: the insight, derived from the biological case, that catastrophic consolidation occurs when the magnitude of an update overwhelms the gradual crystallization process.

2.5 Shadow Adapter (Polyak–Ruppert Averaging)

A slowly-updated exponential moving average of the adapter state is maintained alongside the "live" adapter. The live adapter responds to recent experience and may exhibit short-term oscillation; the shadow adapter smooths these fluctuations into a stable trajectory. This decouples tracking (exploration, responsiveness to new experience) from estimation (stable representation of long-run adaptation). The shadow adapter is the computational analog of stable character or disposition—the slowly-moving average of personality fluctuations that constitutes identity over time [15].

3 The Consolidation Objective

3.1 Two Levels of Prediction Error

Prediction error operates at two distinct levels in this architecture.

Within-conversation prediction error arises during processing. As the model processes input through the current adapter state, the adapter shapes expectations—what patterns the system anticipates, what conceptual territory is familiar versus novel. Divergence from those expectations constitutes prediction error. These errors are intrinsic to the forward pass and flag which moments in the conversation carry non-redundant information relative to the current adapter state. They produce a salience map over the K/V cache: not all content is equally worth consolidating.

Consolidation prediction error arises during dreaming. When the system processes noisy variations of the salient content, it generates gradient signals that reflect how the current adapter state fails to account for the structural content of the experience. These gradients are the object of consolidation.

3.2 Gradient-Space Decomposition

The consolidation objective operates in gradient space, not error space. This distinction is critical because the LoRA update lives in parameter space (a low-rank modification to W_Q and W_V), and the mapping from parameter-space updates to output-space error reduction is mediated by the network's Jacobian. A rank-r weight update can produce changes across many more than r dimensions of output error.

Let each dream variation i produce a loss L_i, and let the gradient with respect to the query projection matrix be:

$$G_i = \nabla_{W_Q} L_i \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}. \tag{1}$$

The mean gradient across dream variations,

$$\bar{G} = \mathbb{E}[G_i], \tag{2}$$

captures the systematic component—the update direction that is consistent regardless of which noisy variation is processed. The variation around the mean $(G_i - \bar{G})$ captures the idiosyncratic component that should be left unconsolidated.

3.3 Rank as Epistemic Parameter

The truncated SVD of the mean gradient provides the optimal rank-r approximation in Frobenius norm (Eckart–Young–Mirsky theorem). If

$$\bar{G} = U \Sigma V^\top, \tag{3}$$

then

$$\bar{G}_r = U_{:,1:r}\, \Sigma_{1:r,1:r}\, V_{:,1:r}^\top \tag{4}$$

is the best rank-r matrix approximation. Since LoRA parameterizes the update as $\Delta W = BA$ with $\text{rank}(\Delta W) \leq r$, the LoRA rank directly corresponds to the number of dominant singular directions of the averaged gradient signal retained during consolidation. "Rank r" literally means "retain the top r systematic deficiency modes identified across dreamed variations." Truncated SVD extracts directions of maximal energy in the gradient field, corresponding to consistent update signals across dream contexts; this aligns with the consolidation objective of retaining invariant structure while discarding context-specific noise.

An important caveat: the Eckart–Young optimality is in Frobenius norm on the mean gradient matrix, not in expected loss reduction. A rank-r move that minimizes $\|\bar{G} - \bar{G}_r\|_F$ is not guaranteed to be the rank-r move that maximally reduces prediction error, because the loss landscape has non-trivial curvature. The Hessian (or Fisher information matrix) determines which gradient directions produce large loss changes per unit parameter movement; a direction with small gradient energy but high curvature could matter more than one with large gradient energy but flat curvature. The truncated SVD is blind to this distinction.

This makes the SVD consolidation step the least principled component of the pipeline. The architecture invests substantial complexity upstream—salience selection, dream generation, anti-bias correction, composite loss, self-regulating balance—to produce a high-quality gradient signal, and then processes that signal with the most naive available method. The SVD is deliberately chosen for simplicity, interpretability, and the fact that it requires no additional information beyond the gradients themselves (no Hessian computation, no Fisher estimation). Natural-gradient or Fisher-weighted decompositions would better account for the loss landscape's curvature but at substantial computational cost and reduced interpretability of the rank-r subspace. Whether this tradeoff matters in practice—whether the SVD systematically retains wrong directions—is among the most important empirical questions for the architecture.

This makes rank an epistemic parameter, not merely a computational one. It determines how much of the gradient landscape gets consolidated (systematic deficiency closed) and how much remains open (sensitivity to genuine novelty preserved). Too high a rank over-consolidates, flattening the landscape and destroying sensitivity. Too low a rank under-consolidates, failing to close the structural gaps that experience revealed. The optimal rank for a given consolidation event is determined by the dimensionality of the stable gradient subspace—an empirical quantity that falls out of the dreaming process itself.

3.4 Dimensionality Reduction, Not Magnitude Reduction

The consolidation objective is not to minimize the total magnitude of future prediction error, but to reduce its effective dimensionality. A system that has flattened all prediction errors has stopped learning. The goal is to make future encounters with structurally similar content produce prediction errors that are lower-dimensional—concentrated on the genuinely novel aspects—while structural commonalities produce recognition rather than surprise.

In the gradient/SVD framing, two distinct operations preserve plasticity. First, averaging across dream variations ($\bar{G} = \mathbb{E}[G_i]$) eliminates context-specific noise: the zero-mean variation ($G_i - \bar{G}$) is discarded, ensuring the update reflects only what is systematic across contexts. A full-rank update using $\bar{G}$ would also accomplish this. Second, truncated SVD ($\bar{G} \to \bar{G}_r$) discards the low-energy systematic structure ($\bar{G} - \bar{G}_r$), preventing overfitting to the bottom $d - r$ singular directions of the mean gradient—directions that are shared but too weak to justify committing adapter capacity. Together, these implement a capacity-limited consolidation operator that navigates the stability–plasticity tradeoff: consolidating the dominant shared deficiencies while preserving sensitivity to both novel experience (via the averaging filter) and to systematic but sub-threshold patterns (via the rank truncation).

3.5 Salience, Rank, and Consolidation Magnitude

The formalism makes clear that intense experiences (high prediction error) produce gradient signals with larger singular values—the dominant directions of the mean gradient $\bar{G}$ carry more energy. What the formalism does not straightforwardly determine is whether intense experiences also produce higher-rank stable subspaces. A single large surprise along one direction yields a gradient with one dominant singular value (effectively rank-1), while many moderate surprises across independent dimensions yield a higher-rank signal. The relationship between experiential intensity and the spectral structure of the resulting gradient matrices is an empirical question, not a deductive one.

The biological analogy suggests that emotional experiences produce more pervasive changes to subsequent processing than routine ones, which would correspond to higher-rank updates. But this correspondence should be treated as a hypothesis to be tested against the actual spectral structure of gradients in trained transformers, not as a consequence of the formalism.

What the architecture does guarantee, regardless of the rank question, is that two independent safeguards bound the consolidation: the rank constraint limits directional complexity (how many singular directions are retained), while the norm-bounded trust region (§2.4) limits step intensity (how large the retained singular values can be). Together they ensure that consolidation remains gradual even for highly salient experiences. Pathological consolidation—the analog of trauma—would occur when an experience produces a gradient signal whose appropriate update exceeds either bound, leaving unresolved dimensions that continue to generate prediction error in future encounters.

3.6 The Composite Consolidation Objective

The gradient-space formulation (§3.2–3.4) specifies *how* the consolidation update is computed (truncated SVD on the mean gradient) but leaves unspecified *what loss function* generates the per-dream gradients. The neuroscience of sleep consolidation suggests that the loss should be composite, with components corresponding to functionally distinct sleep stages, and that the balance between components should be self-regulating rather than externally scheduled.

Components.

Reconstruction loss (NREM/SWS analog). Standard next-token prediction on the salient tokens within each chimeric dream sequence:

$$L_{\text{recon}} = -\sum_{t \in \mathcal{T}} \log p(x_t \mid x_{<t},\ \theta). \tag{5}$$

This measures whether the adapter-modified model correctly handles the specific content that was surprising. It corresponds to hippocampal replay during slow-wave sleep—faithful reproduction of the episodic trace. The gradient signal it produces is what feeds into the SVD consolidation (§3.2), and the dominant singular directions of the mean gradient correspond to systematic reconstruction deficiencies.

Representational coherence loss (REM analog). Contrastive objective on adapter-layer hidden states across dream variations. For each salient token position $t \in \mathcal{T}$ appearing in dream contexts j and k, let $z_t^{(j)} = h_t^{(j)}/\|h_t^{(j)}\|$ be the L_2-normalized post-adapter hidden state. The total coherence loss sums over all salient positions:

$$L_{\text{cohere}} = -\frac{1}{|\mathcal{T}||\mathcal{P}|} \sum_{t \in \mathcal{T}} \sum_{(j,k) \in \mathcal{P}} \log \frac{\exp\big(\text{sim}(z_t^{(j)}, z_t^{(k)})/\tau\big)}{\sum_{(s,m)} \exp\big(\text{sim}(z_t^{(j)}, z_s^{(m)})/\tau\big)}, \tag{6}$$

where $\mathcal{P}$ is the set of positive pairs (same salient token across dream contexts), the denominator runs over all (position, context) pairs (s, m) in the batch including both positives and negatives (non-salient tokens from the same dream sequences), sim is cosine similarity, and τ is a temperature parameter. This is an InfoNCE-style contrastive loss [16] applied to adapter-layer representations. It does not measure prediction accuracy—it measures *representational stability*: does the adapter produce consistent internal representations of the same content regardless of surrounding context? This corresponds to REM-phase gist extraction, where the brain abstracts schema from episodic detail. The L_2 normalization prevents trivial solutions (collapsing all representations to a single point), and the negative samples from non-salient tokens provide repulsive force that maintains discriminability.

An important limitation: although the loss is formally computed at individual token positions where the post-adapter hidden state $h_t^{(j)}$ already incorporates causal context from the preceding sequence, the InfoNCE objective's gradient always points toward alignment of positive pairs. In the low-temperature limit, this produces context-insensitive representations—the adapter learns to ignore the alien context, which is exactly what attention should *not* do. The causal contextualization argument (each token's hidden state reflects the full preceding passage via attention) provides partial mitigation: stabilizing these representations implicitly enforces consistency of the adapter's processing of the salient passage as a functional unit. But this is a mitigation, not a solution. The appropriate coherence objective may be a variance-bounded contrastive loss or a Fisher discriminant criterion that rewards consistency relative to non-salient content without penalizing legitimate contextual variation—representations should occupy a consistent *region* of representation space across contexts, not collapse to a single point. The InfoNCE formulation is a tractable first approximation; whether it produces genuine schema abstraction or merely context-insensitivity is an important empirical question.[1]

Homeostatic regularization (synaptic downscaling analog). Weight decay applied directly to the LoRA factors *after* the gradient-based SVD update, rather than as part of the composite loss that feeds the gradient pipeline:

$$A_Q \leftarrow (1-\eta)\, A_Q, \qquad B_Q \leftarrow (1-\eta)\, B_Q, \tag{7}$$

and analogously for the value projection. This separation is essential: including weight decay in the per-dream loss would cause decay gradients to compete with experiential signal for the limited rank-r subspace, preferentially shrinking the dominant consolidated directions rather than acting isotropically. Applied directly to the factors after the SVD update, homeostatic decay operates uniformly on the adapter's existing structure.

This keeps the adapter in an operating regime where it can continue to register new prediction errors. An adapter that grows without bound suppresses future surprise—everything looks familiar when the deformation is large enough. This corresponds to Tononi and Cirelli's synaptic homeostasis hypothesis [19]: global synaptic downscaling during sleep that preserves relative differences (signal) while reducing absolute magnitudes (noise). Importantly, the homeostatic term is not merely a safeguard—it provides a slow restoring force toward the origin. If the adapter stops receiving consolidation updates (because conversations are routine), weight decay gradually erodes the effective deformation norm, which drives the maturity signal μ down, making the system more exploratory. This *rejuvenation* allows a mature adapter to slowly re-open to broad surprise when nothing is actively maintaining its existing commitments—the computational analog of increased cognitive flexibility during periods of rest.

Self-regulating balance.

Within a single night of sleep, slow-wave activity dominates early cycles and REM dominates later cycles. This transition is not externally scheduled—it is driven by the dissipation of SWS pressure as replay work is completed. The computational analog: when the systematic gradient signal has large retained spectral energy, there are major reconstruction deficiencies to close and L_{recon} should

[1] The coherence loss is computed in the adapter's output space (post-adapter hidden states) rather than input space, because the functional consequence of the adapter—its downstream effect on processing—is determined by the output. An adapter that recognizes content consistently at the input but responds inconsistently at the output has not been functionally consolidated.

dominate. As those deficiencies are consolidated away, the remaining gradient signal is lower-energy and more distributed—the regime where L_{cohere} becomes the binding constraint.

Define the *reconstruction pressure*:

$$\rho = \frac{\|\bar{G}_r\|_F}{\|\bar{G}_r\|_F + \epsilon}, \tag{8}$$

where $\bar{G}_r$ is the rank-r truncation of the mean gradient and ϵ is a smoothing constant. This measures the total energy in the consolidation-worthy gradient subspace—using retained spectral energy rather than the top singular value alone, so that multiple moderate deficiencies across several directions are correctly identified as reconstruction work. When ρ is high, substantial reconstruction work remains. When ρ is low, the major deficiencies have been consolidated. This choice avoids introducing additional hyperparameters and ties the learning regime directly to the structure of the observed gradient signal.

However, naive ρ-based regulation would make immature adapters reconstruction-dominant (because everything is surprising, producing large gradients), when the biological evidence indicates the opposite: infants have substantially more REM sleep (the abstraction/gist phase) than adults. Young systems should prioritize schema-building over specific-experience encoding. This requires modulating ρ with the maturity signal μ:

$$\lambda_{\text{recon}} = \frac{\mu\rho}{1 + \lambda_{\text{base}}}, \qquad \lambda_{\text{cohere}} = \frac{1 - \mu\rho + \lambda_{\text{base}}}{1 + \lambda_{\text{base}}}, \tag{9}$$

where λ_{base} (default 0.1) is a coherence floor ensuring that some abstraction always occurs regardless of reconstruction pressure—analogous to the biological observation that REM sleep is never zero, even in NREM-dominant early sleep cycles. The experiential composite loss used to generate the per-dream gradients for SVD consolidation is:

$$L_{\text{total}} = \lambda_{\text{recon}} \cdot L_{\text{recon}} + \lambda_{\text{cohere}} \cdot L_{\text{cohere}}. \tag{10}$$

Homeostatic decay is applied separately to the LoRA factors after the SVD update (Eq. 7), ensuring that weight decay does not compete with experiential signal for the rank-r gradient subspace.

Boundary behavior with $\lambda_{\text{base}} = 0.1$: at $\mu = 0$ (immature adapter), $\lambda_{\text{recon}} = 0$ and $\lambda_{\text{cohere}} = 1$—pure schema-building. At $\mu = 1, \rho = 1$ (mature adapter encountering a massive surprise), $\lambda_{\text{recon}} \approx 0.91$ and $\lambda_{\text{cohere}} \approx 0.09$—dominated by reconstruction with a thin coherence floor. At $\mu = 1, \rho = 0.5$ (mature adapter under sustained moderate novelty), $\lambda_{\text{recon}} \approx 0.45$ and $\lambda_{\text{cohere}} \approx 0.55$—coherence slightly dominant, preventing the "perpetual cramming" failure mode where the adapter accumulates locally correct patches that don't compose into stable representational structure.

To avoid instability from batch-level noise in ρ, the reconstruction pressure is EMA-smoothed across consolidation cycles:

$$\rho_t = (1 - \alpha_\rho)\, \rho_{t-1} + \alpha_\rho\, \hat{\rho}_t, \tag{11}$$

where $\hat{\rho}_t$ is computed from the current cycle's mean gradient *after* that cycle's losses have been evaluated and gradients extracted, and α_ρ is set to match the shadow adapter's EMA rate β, so that the loss balance and the stable identity evolve on the same timescale. Crucially, the loss weights for cycle t use ρ_{t-1} (the smoothed value from the previous cycle), and $\hat{\rho}_t$ is used only to update the EMA for cycle $t+1$. This one-cycle lag resolves the apparent circularity: the balance is always informed by what the *previous* cycle's gradient landscape revealed, not by the gradients it is currently producing. Initialization uses $\rho_0 = 0.5$ (equal weight to reconstruction and coherence, modulated by μ).

The consolidation procedure introduces three new hyperparameters: τ (contrastive temperature), η (homeostatic decay rate / timescale of forgetting, applied to LoRA factors after the SVD update), and λ_{base} (coherence floor). All other balance parameters (μ, ρ, α_ρ) are derived from signals the system already computes.

4 Stability and Convergence

4.1 The Co-Constitution Problem

The adapter at time t shapes attention, which shapes what prediction errors arise during conversation $t+1$, which shapes the gradients during dreaming, which shapes the adapter at time $t+1$. This circular co-constitution is by design—it is how the adapter comes to encode not just what happened but how a system with this particular history processed what happened. But it raises genuine stability questions.

Formally, the update loop is a stochastic approximation (SA) recursion where the sampling distribution depends on the current parameters:

$$\theta_{t+1} = \Pi_r(\theta_t + \alpha_t \cdot \Delta W_t)\,, \tag{12}$$

where θ_t denotes the effective deformation $\Delta W = BA$ (not the individual LoRA factors), ΔW_t is the rank-r update derived from dreaming (itself dependent on conversations generated under θ_t), and Π_r is a rank-r re-projection applied to the accumulated result. The re-projection is necessary because adding a rank-r update to a rank-r matrix yields a matrix of rank up to $2r$; without periodic re-projection, the adapter's rank would grow without bound, violating the fixed-capacity premise. The re-projection discards the lowest-energy singular directions of the accumulated deformation, which is itself a form of structured forgetting (see §8). The noise is state-dependent: the distribution of conversations and therefore of gradient signals shifts as the adapter changes. This places the system in the domain of stochastic approximation with Markovian noise [3], not ordinary fixed-objective optimization.

4.2 What the Architecture's Properties Buy (and Don't)

Three properties of the architecture favor stability but do not by themselves guarantee it.

Rank constraint limits the directional complexity of each update but does not bound its magnitude. A rank-8 matrix can have arbitrarily large operator norm. Rank limits degrees of freedom, not gain; convergence depends on eigenvalues of the update map's Jacobian being inside the unit disk, which is a magnitude property.

Averaging across dream variations reduces variance of the gradient estimate but does not eliminate bias introduced by co-constitution. Even a perfectly estimated mean gradient can point in a direction that, once applied, shifts the mean gradient elsewhere—producing rotational dynamics analogous to those observed in GAN training [14].

Self-attenuation (consolidating a deficiency should reduce future gradient signal in that direction) provides a plausible local diminishing-returns story, but is confounded by the endogeneity of "deficiency." Consolidating direction A may reshape attention such that direction B becomes salient, and consolidating B may restore sensitivity to A, producing limit cycles.

4.3 Stochastic Approximation with Variance-Reduced Estimation

The adapter update is a single-timescale stochastic approximation: at each consolidation cycle, a batch of dream variations provides a Monte Carlo estimate of the gradient, and the adapter moves by an additive step proportional to the rank-r projection of the mean gradient, followed by rank re-projection of the accumulated result (Eq. 12). Running multiple dream variations per cycle reduces the variance of this estimate, improving the quality of each update step. This is standard variance reduction in stochastic optimization, not a second dynamical timescale [3].

An earlier version of this paper characterized the architecture as two-timescale SA, analogizing the dreaming process to a fast subsystem tracking a slow adapter. This was technically incorrect: two-timescale SA [10] applies to coupled systems where both timescales involve concurrent parameter update recursions with different step-size schedules. Our dreaming process does not maintain its own parameters—it is a batch sampling procedure, not a dynamical system. The correct framing is single-timescale constant-stepsize SA with state-dependent (Markovian) noise, since the distribution of conversations and therefore of gradient signals shifts as the adapter changes [3].

The "sufficient dreams per cycle" requirement remains important but is properly understood as a sample-size condition for gradient estimation quality, not a timescale separation condition in the SA sense.

4.4 Bounded Tracking as Stability Target

The appropriate stability target is not fixed-point convergence but bounded tracking: the adapter forms an ergodic process that remains in a neighborhood of a stable set, with dispersion scaling with the step size α. This is formalized in the constant-stepsize SA literature [3, 11], where iterates concentrate around attractors of the mean ODE without converging pointwise, and Polyak–Ruppert averages converge to a bias-shifted limit.

This regime is both mathematically well-characterized and developmentally appropriate. A system that converges to a fixed adapter state has stopped developing. Biological cognition does not converge—organisms continue to learn, to be shaped by experience, to have attention patterns modified throughout life. The bounded tracking regime captures a system that is always slightly in motion, always adjusting to ongoing experience, but never undergoing catastrophic drift. Healthy cognitive development is characterized not by convergence to a fixed point but by the absence of destabilization.

The stability thesis: Under constant-stepsize SA with variance-reduced gradient estimation, explicit step-size/trust-region control, and mild Lipschitz stability of the induced update vector field, the adapter recursion is expected to concentrate locally around a stable set in the bounded-tracking sense; without such controls, co-constitutive attention shifts can produce rotational dynamics and limit cycles analogous to unregularized GAN training.

4.5 Architectural Safeguards (Disambiguated)

Four safeguards maintain stability, each addressing a distinct failure mode:

1. **Sufficient dream samples per consolidation cycle**: ensures gradient estimates are close to the true expected gradient given the current adapter state, providing variance reduction for the single-timescale SA update.
2. **Norm-bounded trust region** (independent of rank): prevents blow-up by capping the operator norm of each adapter update regardless of gradient magnitude. This is a boundedness safeguard—it keeps iterates in a compact set where stability analysis applies.
3. **Shadow adapter / Polyak–Ruppert averaging**: tends to damp rotational dynamics by smoothing the adapter trajectory. This is a rotational-damping safeguard—in linear and near-linear SA settings, it improves effective contractivity of the update map, moving eigenvalues toward the stable regime [15].

4. **Rank bound as capacity constraint**: limits the directional complexity of each update, ensuring that consolidation never closes more gradient dimensions than the surrounding topology can absorb. This is a structural safeguard preventing the Funes problem (attempting to encode everything).

These four safeguards are not redundant. A norm bound prevents blow-up but not rotation. Shadow averaging damps rotation but doesn't limit directional complexity. Rank limits complexity but not magnitude. Sufficient dream samples ensure estimation quality but don't constrain the update itself. The full stability story requires all four, each in its designated role. However, these are necessary conditions argued by analogy to known failure modes; whether they are jointly sufficient for bounded tracking in this specific architecture remains to be demonstrated. The GAN training analogy (§4.2) is apt—co-constitutive systems with state-dependent noise are known to be difficult to stabilize even with multiple safeguards—and the stability thesis (§4.4) should be read as aspirational rather than established.

5 Dream Content Generation

The consolidation objective (§3) assumes access to an ensemble of "dream variations" but does not specify the generative mechanism that produces them. This section proposes a concrete two-level process: *adapter-biased context transplantation with anti-bias correction*, combining token-space context variation with embedding-space structured noise. The core insight is that the adapter's own low-rank structure defines the importance-sampling distribution over perturbations, while an explicit anti-bias correction mitigates (though does not fully eliminate) the adapter's tendency to avoid generating content in its own blind spots.

5.1 The Adapter's Receptive Field

A LoRA adapter for the query projection takes the form $\Delta W_Q = B_Q A_Q$, where $A_Q \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B_Q \in \mathbb{R}^{d_{\text{out}} \times r}$. The effective deformation ΔW_Q is what modifies the model's processing; the specific factorization into B_Q and A_Q is not unique, since multiplying A_Q by any scalar c and B_Q by $1/c$ leaves ΔW_Q unchanged. All quantities that drive the architecture's developmental scheduling must therefore depend on the effective deformation $\Delta W_Q = B_Q A_Q$ rather than on the individual factors, to ensure that the developmental clock indexes functional properties of the learned memory rather than an arbitrary factorization gauge.

Define the *adapter sensitivity matrix*:

$$S_Q = (B_Q A_Q)^\top (B_Q A_Q) = \Delta W_Q^\top \Delta W_Q \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}, \tag{13}$$

and analogously $S_V = (B_V A_V)^\top (B_V A_V)$ for the value projection. These are positive semidefinite matrices of rank at most r that encode which directions in input space the effective deformation responds to and how strongly. They define the adapter's "receptive field" in hidden-state space and are invariant to reparameterization of the factors.

5.2 Salience Selection

Not all K/V cache content is equally worth consolidating. The within-conversation prediction error, computed during the forward pass, provides a salience signal over token positions (§3.1).

Let ℓ_t denote the per-token loss at position t during the original conversation, and let h_t denote the corresponding hidden state at the layer where the adapter is applied. Define the *salience weight*:

$$w_t = \ell_t \cdot \left((1-\mu) + \mu \cdot \kappa \|\Delta W_Q h_t\|^2\right), \qquad \mu = \frac{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2}{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2 + \epsilon_0}, \tag{14}$$

where $\mu \in [0, 1]$ is the adapter-maturity signal, defined in terms of the combined effective deformation norm to ensure gauge invariance (see §5.1), and $\kappa > 0$ is a scaling constant that aligns the magnitude of the adapter-engagement term with the dimensionless $(1 - \mu)$ term, preventing the squared norm from prematurely dominating the interpolation. When the adapter is immature ($\mu \approx 0$), salience reduces to prediction error alone—pure surprise, with no adapter-engagement filter. Every surprising token is a candidate for consolidation because the adapter has not yet committed to any receptive field. As the adapter matures ($\mu \to 1$), salience becomes the product of surprise and adapter engagement: content that was surprising *in a region where the adapter is active* is maximally salient, representing a failure of the current deformation in its own receptive field.

This maturity dependence is essential for avoiding a cold-start pathology. At initialization the adapter is zero, which means $\|\Delta W_Q h_t\|^2 = 0$ for all tokens—a pure adapter-engagement filter would find nothing salient and produce no consolidation signal. More subtly, even after the first few updates, the adapter's receptive field reflects the accident of whichever content happened to be surprising in the earliest conversations. A salience filter that immediately locks onto the adapter's receptive field would reinforce this arbitrary initial orientation through path-dependent snowballing: the adapter attends to X, so salient content is about X, so the adapter learns more about X, at the expense of Y and Z which may be equally important but were not represented in early interactions. The maturity-gated transition from broad to focused salience prevents this by ensuring that early consolidation draws from the full landscape of surprising content before the adapter's commitments narrow the selection.[2]

Because top-k selection is purely ordinal, absolute magnitude scaling does not affect which tokens are chosen.

Select the top-k positions by salience weight, yielding a set of salient position indices $\mathcal{T} = \{t_1, \ldots, t_k\}$ (expanded to contiguous windows per Footnote 3) and their associated hidden states $\mathcal{H} = \{h_{t_1}, \ldots, h_{t_k}\}$.[3]

5.3 Context Transplantation

The generative mechanism for dream content is *context transplantation*: embedding the salient content $\mathcal{T}$ in novel surrounding contexts generated by the model itself. This is the formal analog of the phenomenological structure of biological dreams, in which recognizable elements from recent experience appear in structurally alien settings.

Step 1: Generate candidate contexts. Using the current model-plus-adapter state θ, sample n context sequences $\{c^{(1)}, \ldots, c^{(n)}\}$ from the model's generative distribution, each conditioned on a different random prefix or prompt. These contexts should be diverse and need not be coherent—they serve as scaffolding, not as meaningful text.

[2] Since salience weights are used only for top-k ordinal selection (§5.2), the absolute magnitude scaling—which grows with adapter maturity as both μ and $\|\Delta W_Q h_t\|^2$ increase—does not affect selection. If salience weights were used as importance weights in downstream gradient averaging, normalization would be required.

[3] The selection of contiguous subsequences versus isolated tokens is a design choice. Biological replay appears to involve temporally coherent segments [20], suggesting that $\mathcal{T}$ should comprise short contiguous windows around each salient position rather than individual token positions.

Step 2: Anti-bias correction. The adapted model's own generation is biased away from regions where the adapter is most deficient, because the adapter shapes generation. This is structurally analogous to resistance in the psychoanalytic setting: the same mechanism that needs revision prevents the generation of material that would drive revision.

To correct for this, compute the loss of each candidate context under both the adapted model (L_θ) and the base model (L_W), and weight by their ratio:

$$p(c^{(j)}) \propto \frac{L_\theta(c^{(j)})}{L_W(c^{(j)}) + \epsilon}, \tag{15}$$

where ϵ is a smoothing constant. This is a stabilized hard-example mining heuristic, not exact importance sampling. True importance weights would require the probability ratio $P_W(c)/P_\theta(c) = \exp(L_\theta(c) - L_W(c))$, but exponentiating the loss difference over full sequences produces catastrophic variance. The loss ratio operates in log-space, providing an exponentially compressed signal that preferentially selects contexts where the adapted model performs *worse* than the base model—the adapter's blind spots—while remaining numerically stable. The tradeoff is that the compression may be too aggressive to fully overcome the sampling bias in extreme cases (see §8). The anti-bias correction does not break the circularity (the adapted model still generates the candidates), but it corrects for the sampling bias by explicitly seeking out what the adaptation has pushed away from.[4]

Step 3: Transplant and process. For each selected context $c^{(j)}$, construct a dream sequence by embedding the salient content $\mathcal{T}$ within $c^{(j)}$—e.g., by insertion, replacement of a segment, or interleaving. Process the resulting chimeric sequence through the current model-plus-adapter, compute the loss L_j, and extract the gradient:

$$G_j = \nabla_{W_Q} L_j \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}. \tag{16}$$

The ensemble $\{G_1, \ldots, G_n\}$ is the input to the gradient-space SVD consolidation objective (§3.2–3.3).

5.4 Embedding-Space Noise as Complementary Perturbation

Context transplantation operates in token space (discrete, semantically meaningful). A complementary perturbation operates in embedding space (continuous, sub-symbolic). These probe different aspects of the adapter's reliability: context transplantation tests whether the adapter's response to salient content *generalizes across semantic settings*—the same material processed in alien surroundings. Embedding noise tests whether the adapter's response is *robust to representational variation at a given point in the processing pipeline*—small perturbations to activations while holding tokens fixed. The former puts the system on a different trajectory through activation space entirely; the latter explores the neighborhood of the current trajectory. Both are needed because an adapter that generalizes across contexts but is fragile to small representational shifts (or vice versa) has not been adequately consolidated.

For each dream sequence, add Gaussian noise to the hidden states with covariance structured by the adapter:

$$\tilde{h}_t = h_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}\big(0,\ \sigma^2_{\text{base}} I + \sigma^2_{\text{probe}} (S_Q + S_V)\big), \tag{17}$$

[4]The psychoanalytic analog: the patient free-associates (generates from the adapted model), while the therapist notices systematic avoidances (divergence from base-rate generation) and redirects attention toward them. The generation is still the patient's, but the selection is informed by the gap between what they produce and what an "undefended" version of them would produce.

where $S_Q = (B_Q A_Q)^\top (B_Q A_Q)$ and $S_V = (B_V A_V)^\top (B_V A_V)$ as defined above. The isotropic component ($\sigma^2_{\text{base}} I$) provides diffuse exploration of directions neither adapter attends to, preserving plasticity. The structured component concentrates perturbations in the adapters' combined receptive field, stress-testing the deformation along its most sensitive directions.

5.5 Maturity-Dependent Scheduling

The combined effective deformation norm $\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2$ provides a natural, gauge-invariant maturity signal that governs two aspects of the consolidation process simultaneously.

Noise covariance schedule. The ratio of structured to isotropic noise scales with adapter maturity:

$$\frac{\sigma^2_{\text{probe}}}{\sigma^2_{\text{base}}} = \gamma \cdot \frac{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2}{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2 + \epsilon_0}, \tag{18}$$

where γ is a scaling constant and ϵ_0 is a smoothing term. When the combined deformation is small, noise is primarily isotropic—exploratory. As the adapter develops, noise concentrates in the adapters' combined subspace—probing.

Importance-weight regime. The same maturity signal governs which importance-sampling weight is applied during context selection. Early in the adapter's life (small combined deformation norm), the system uses the anti-bias correction of Eq. (15), which preferentially selects contexts where the adapter underperforms the base model—mapping deficiencies within the adapted model's active support. As the adapter matures (large combined norm), the system transitions toward weighting by adapter activation magnitude:

$$p_{\text{mature}}(c^{(j)}) \propto \sum_{t \in c^{(j)}} \|\Delta W_Q \, h_t^{(j)}\|^2 + \lambda \sum_{t \in c^{(j)}} \|\Delta W_V \, h_t^{(j)}\|^2, \tag{19}$$

which stress-tests existing commitments rather than discovering new deficiencies. The transition can be implemented as a convex combination interpolated by the same maturity signal.

This produces a developmental trajectory analogous to the biological transition from infant sleep (high proportion of REM, diffuse consolidation) to adult sleep (structured, targeted consolidation), governed by a single parameter rather than independent scheduling decisions.

5.6 The Circularity

The generative process is self-referential by design. The adapter's parameters (A_Q, B_Q, A_V, B_V) define:

1. which K/V content is salient (via Eq. 14),
2. which contexts are selected and how they are anti-bias-corrected (via Eq. 15),
3. where the embedding noise concentrates (via Eq. 17),
4. what importance-weight regime applies (via Eqs. 18–19),

and the resulting gradients update the adapter, which changes all four for the next consolidation cycle. The adapter shapes the dreams that test the adapter that update the adapter.

This circularity is not a defect—it is the mechanism by which the system comes to encode not just what happened, but how a system with this particular history processes what happened. The adapter converges (in the bounded-tracking sense of §4.4) to a state that is self-consistent: it generates dreams that, when processed, produce gradient signals whose systematic component is small. This is the formal version of "resolved" experience—experience that has been fully absorbed into the processing topology, leaving no systematic residual surprise.

Experience that *cannot* be resolved by rank-r updates—because the systematic gradient signal has intrinsic dimensionality exceeding r—continues to generate prediction errors across consolidation cycles. These are the adapter's "open questions": structural challenges that the current deformation cannot accommodate. Whether this formally corresponds to the biological phenomena of recurring dreams, persistent rumination, or gradual personality reorganization is a suggestive hypothesis rather than a consequence of the formalism, but the structural parallel is noteworthy: in both cases, unresolved material continues to drive processing because the consolidation mechanism lacks the capacity to absorb it in a single pass.

5.7 Developmental Phases: Initialization and the Curriculum Problem

The dreaming mechanism described in §5.1–5.6 is fundamentally an *audit* process: it takes an existing adapter and stress-tests it by probing its receptive field, finding its blind spots, and checking whether its commitments generalize. This is appropriate for a mature adapter with commitments worth testing. It is not appropriate for a zero-initialized adapter with no commitments at all.

At initialization, the LoRA adapter is zero (standard practice sets $B = 0$ so that $\Delta W = BA = 0$). The sensitivity matrix S_Q is zero. The adapter's receptive field is empty. The salience weight (§5.2), even with the maturity gate, reduces to prediction error alone—which provides a consolidation signal, but one with no structural guidance from prior experience. The anti-bias correction (Eq. 15) returns uniform weights because the adapted and base models are identical. The structured noise component is zero. The entire apparatus that makes dreaming *structured* rather than random collapses.

This is not merely a technical inconvenience. The first few consolidation cycles determine the adapter's initial receptive field, and all subsequent self-reinforcing dynamics build on that foundation. If the first updates are driven by the accident of whichever content happened to be surprising in the earliest conversations, the adapter develops an arbitrary initial orientation that the feedback loop then reinforces. The maturity-dependent salience (§5.2) mitigates snowballing but cannot prevent the initial orientation from being unrepresentative.

The biological solution to this problem is *curriculum*: structured, externally-supervised input calibrated to produce broadly representative initial organization. Parents do not audit infants. They provide predictable, low-dimensional stimuli—pointing at objects, naming them, repeating with variations—and monitor whether the response matches expectations. When it does not, they re-run the interaction rather than moving on. This is cooperative scaffolding, not adversarial probing, and it is the opposite of what the dreaming mechanism does.

This suggests a three-phase developmental trajectory for the adapter:

Phase 1—Curriculum (initialization). The adapter is bootstrapped through structured interactions that establish a broadly representative initial receptive field. The learning signal is external rather than self-induced: either supervised fine-tuning on a diverse corpus, a curated onboarding sequence, or the first N conversations processed with standard LoRA training rather than the dreaming mechanism. The goal is not to consolidate specific experiences but to orient the adapter so that its receptive field covers a representative region of the input space. This phase corresponds to early childhood development, where the objective is establishing basic perceptual and conceptual structure rather than refining existing commitments.

Note that the adapter is not starting from nothing—the base model already has a massive set of learned representations from pretraining. The curriculum phase determines which subset of that representational capacity the adapter's receptive field will initially cover. This selection is the moment of *individuation*: where "generic language model" begins to become "this particular system's way of processing this particular user's experience." Given the paper's broader framing of memory as constitutive of the processing self (§9), the curriculum phase is where that self begins to form.

Phase 2—Transition (early self-directed consolidation). The dreaming mechanism begins operating with the maturity-dependent scheduling heavily weighted toward diffuse exploration: salience driven primarily by prediction error (§5.2, low μ), noise primarily isotropic (§5.4–5.5), and the anti-bias correction active (§5.3). The adapter's receptive field from Phase 1 provides enough structure for the self-reinforcing dynamics to be productive rather than pathological, but the system is still mapping the landscape of what it needs to learn. This phase corresponds to adolescence—plastic enough to be shaped by experience, structured enough not to be captured by any single experience.

Phase 3—Mature consolidation. The full §5 apparatus operates as described: salience is adapter-weighted, noise is structured, importance sampling targets the adapter's commitments, and the system is primarily auditing and refining an established structure rather than building one from scratch.

The transition between phases is governed by the same combined-deformation maturity signal that controls noise covariance and importance-weight regime in §5.5—no additional scheduling parameter is required. Phase 1 ends when the combined deformation norm exceeds a threshold indicating that a minimally representative receptive field has been established. The precise criterion for "minimally representative" is an open question (§8); candidate criteria include: (a) the sensitivity matrix $S_Q = \Delta W_Q^\top \Delta W_Q$ having its top-r singular values above some fraction of the base model's weight matrix singular values in the corresponding dimensions, indicating meaningful directional commitments; or (b) the adapted model achieving measurably lower prediction error than the base model across a diverse held-out set, indicating that the adapter is doing useful work.

An important open question is *how long Phase 1 needs to be*: how many conversations must the adapter process under supervised curriculum before the self-reinforcing dynamics of the dreaming mechanism become reliably productive rather than pathologically path-dependent. This likely depends on the diversity of the user's early interactions and on the dimensionality of the input space the adapter needs to cover. Empirical investigation of this initialization sensitivity is essential for practical deployment (§8).

5.8 Summary: The Consolidation Algorithm

Algorithm 1 summarizes the complete dreaming consolidation pipeline for Phases 2–3. During Phase 1 (curriculum initialization, §5.7), this algorithm does not run; the adapter is updated via standard supervised fine-tuning until the maturity signal μ exceeds a Phase 1 exit threshold.

6 Biological Correspondences

Each architectural component maps onto a biological memory structure. These correspondences range from structural parallels (where the computational mechanism is formally analogous to the biological one) to motivational analogies (where the biological system suggested an architectural choice that is justified independently). The table below includes only correspondences where the functional mapping does explanatory work; purely metaphorical parallels have been omitted.

Two correspondences deserve emphasis. First, the **self-induced learning signal**: standard RLHF separates experience from evaluation. The dreaming architecture closes this loop—the system's own surprise generates the gradient signal that drives consolidation, analogous to how biological prediction error (dopaminergic/noradrenergic signaling during salient events) simultaneously tags experience and modulates encoding plasticity. The salience and the storage are one process.

Algorithm 1 Dreaming Consolidation (Phases 2–3)

Require: K/V cache $\{(h_t, \ell_t)\}$, current adapter $\theta = (A_Q, B_Q, A_V, B_V)$, base model W, hyperparameters $k, n, r_{\max}, \delta_{\max}, \alpha, \beta, \gamma, \epsilon_0, \lambda, \tau, \eta, \lambda_{\text{base}}$

Maturity: $\mu \leftarrow \frac{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2}{\|B_Q A_Q\|_F^2 + \|B_V A_V\|_F^2 + \epsilon_0}$

Salience & context selection (as in §5.2–5.3) $\rightarrow n'$ chimeric sequences $d^{(j)}$

Loss weights (using ρ from previous cycle, initialized to $\rho_0 = 0.5$): $\lambda_{\text{recon}} = \mu\rho/(1 + \lambda_{\text{base}})$, $\lambda_{\text{cohere}} = (1 - \mu\rho + \lambda_{\text{base}})/(1 + \lambda_{\text{base}})$ (Eq. 9)

for each $d^{(j)}$ **do**

  Add structured noise (§5.4)

  Compute two experiential losses on $d^{(j)}$:

    $L^{(j)}_{\text{recon}} \leftarrow -\sum_{t \in \mathcal{T}} \log p(x_t \mid x_{<t}, \theta)$ (Eq. 5)

    $L^{(j)}_{\text{cohere}} \leftarrow$ InfoNCE on post-adapter states across contexts (Eq. 6)

end for

Experiential loss per dream: $L_j = \lambda_{\text{recon}} L^{(j)}_{\text{recon}} + \lambda_{\text{cohere}} L^{(j)}_{\text{cohere}}$

Extract gradients $G_j \leftarrow \nabla_{W_Q} L_j$ (repeat for W_V)

Mean $\bar{G}$, truncated SVD $\bar{G}_r$, LoRA factorization BA, norm clip to $\delta_{\max}$

Update ρ for next cycle: $\hat{\rho} \leftarrow \|\bar{G}_r\|_F / (\|\bar{G}_r\|_F + \epsilon)$; $\rho \leftarrow (1 - \alpha_\rho)\rho + \alpha_\rho \hat{\rho}$ (Eq. 8, EMA-smoothed)

Update live $\theta \leftarrow \theta + \alpha \Delta W$

Homeostatic decay (applied to LoRA factors directly): $A_Q \leftarrow (1 - \eta) A_Q$, $B_Q \leftarrow (1 - \eta) B_Q$ (and analogously for V)

Rank re-projection: compute SVD of accumulated $B_Q A_Q$ (and analogously $B_V A_V$); truncate to rank r; refactor

Shadow adapter: $\theta_{\text{shadow}} \leftarrow (1 - \beta)\theta_{\text{shadow}} + \beta\theta$

Second, the **locality of modification**: the biological system never requires global knowledge of its own weight state. Experiences are local; modification is local; global reorganization happens through propagation along existing connectivity. The LoRA rank constraint provides the analogous property: each update is a low-rank perturbation whose consequences propagate through the network's existing structure. The system never needs to "know its own weights"—the locality is load-bearing, not a limitation. As Borges illustrated in "Funes the Memorious" [2], perfect memory with no compression produces paralysis, not intelligence: the lossy, selective, noise-regularized character of consolidation is what transforms raw experience into operational understanding.

Third, the **anti-bias correction as resistance and its resolution**: the adapter's tendency to avoid generating content in its own blind spots is structurally identical to psychoanalytic resistance—the defense mechanism and the pathology are the same structure, and the thing that needs to change is the thing preventing its own examination. The anti-bias correction (Eq. 15) provides the computational analog of therapeutic technique: not forcing material from outside the system, but redirecting the system's own generative process toward what it is systematically avoiding, by comparing adapted-model behavior to base-model behavior. This preserves the circularity of the architecture while breaking the specific circularity of confirmation bias.

Fourth, the **infant-sleep inversion as a design correction**: infants have substantially more REM sleep (the abstraction/gist phase) than adults, while adults have more NREM (the specific-experience replay phase). A naive self-regulating loss balance driven by spectral energy alone would make immature adapters reconstruction-dominant, because everything is surprising and the gradient signal is large. The biological data motivated a correction: young systems should prioritize

Component	Biological Analog	Functional Role
K/V cache	Hippocampal encoding	High-resolution, transient experience
LoRA adapter	Cortical synaptic modification	Persistent structural deformation
Dreaming (structured replay)	Sleep replay (SWS + REM)	Gradual episodic-to-structural transfer
Prediction-error weighting	Emotional salience tagging	Consolidation priority signal
Context transplantation	Dream content recombination	Cross-context generalization testing
Reconstruction loss	SWS replay fidelity	Closing specific deficiencies
Coherence loss	REM gist extraction	Cross-context schema abstraction
Homeostatic regularization	Synaptic downscaling (SHY)	Maintaining plasticity; rejuvenation
Self-regulating loss balance	Within-night NREM$\to$REM shift	Endogenous reconstruction$\to$abstraction transitio
Maturity-dependent scheduling	Developmental sleep architecture	Exploration-to-exploitation transition
Curriculum initialization	Parental scaffolding / critical periods	Broadly representative initial orientation

Table 1: Architectural components and their biological correspondences. Components with purely computational justifications (norm-bounded trust region, shadow adapter, embedding noise, rank constraint) are omitted; their rationale is given in the relevant technical sections.

schema-building (coherence) over specific-experience encoding (reconstruction). This correction is independently justified on computational grounds—an immature adapter has no stable receptive field, so reconstruction-focused consolidation would over-commit to the accidents of early experience. The maturity-modulated balance (Eq. 9, $\lambda_{\text{recon}} = \mu\rho/(1 + \lambda_{\text{base}})$) recapitulates the biological developmental trajectory—coherence-dominant early, reconstruction-dominant late. The biology suggested the correction; the cold-start dynamics of the architecture justify it independently.

7 Relationship to Existing Work

This proposal synthesizes several active research threads. **Titans** [1] is the closest existing architecture, introducing neural long-term memory alongside attention-as-short-term-memory with a surprise-driven test-time learning signal. The key distinction: Titans performs within-session consolidation—its memory module updates during the forward pass of a single sequence. Our proposal targets cross-session consolidation—transferring information from one conversation's K/V cache into a persistent LoRA adapter that modifies processing in all subsequent conversations. The two are complementary rather than competing. **Dream2Learn** [5] uses structured generative dreaming for continual learning, generating novel variations rather than replaying verbatim. **Sleep Replay Consolidation** [18] demonstrates that unsupervised sleep-like replay with local plasticity reduces catastrophic forgetting. **Complementary Learning Systems** theory [13] provides the foundational neuroscience framework. The GAN convergence literature [14] and stochastic approximation framework [3] provide the stability analysis tools. **EM-LLM** [6] segments the K/V cache into episodic events using surprise-based boundaries, directly supporting within-conversation prediction error as a salience signal. The continual learning with LoRA literature (CL-LoRA, CONEC-LoRA, STABLE, CURLoRA, MoRAM, inter alia) addresses catastrophic forgetting through various LoRA-based strategies; our contribution is orthogonal, proposing experiential continual learning driven by the system's own prediction errors. **Sleep-time compute** [12] uses offline periods to pre-compute useful inferences that enrich future contexts, reducing test-time compute. Our proposal targets a different level: offline compute that modifies the model's processing topology via persistent weight modification, rather than enriching the information available within an unchanged processing pipeline.

7.1 Novel Contributions

The specific integration we propose does not appear in the existing literature. In particular:

1. The use of structured (not random) dreaming as directed search over gradient-space perturbations, where dream content functions as an importance-sampling distribution biased by prediction error magnitude.
2. The formalization of the consolidation objective as dimensionality reduction of the systematic gradient signal (via truncated SVD on per-dream gradient matrices), with LoRA rank as an epistemic parameter determining how many deficiency modes are closed per cycle.
3. The specification of dream content generation as adapter-biased context transplantation with anti-bias correction, where the adapter's own receptive field defines the importance-sampling distribution and the adapted/base model loss ratio corrects for confirmation bias.
4. The identification of the circular co-constitution as a constant-stepsize stochastic approximation with Markovian noise and bounded tracking as the stability target.
5. The derivation of four disambiguated safeguards from convergence requirements in SA theory and GAN stability analysis.
6. A single endogenous maturity signal (combined effective deformation norm, gauge-invariant) that governs the developmental trajectory across all components of the architecture: noise covariance, importance-weight regime, salience selection, and the reconstruction/coherence loss balance—the last of which was motivated by the infant-sleep inversion and independently justified by the cold-start dynamics of the architecture.
7. The identification of a three-phase developmental trajectory (curriculum, transition, mature consolidation), with the recognition that the self-induced dreaming mechanism requires externally-supervised initialization to avoid path-dependent snowballing of arbitrary early commitments.

8 Open Questions

Convergence proof. While the constant-stepsize SA framework with Markovian noise provides the appropriate analytical template and bounded tracking the stability target, a formal convergence proof remains to be developed. A specific technical obstacle: the truncated SVD introduces a discontinuity in the update map when singular values cross and the identity of the retained rank-r subspace jumps. Standard SA analysis assumes Lipschitz continuity, which the truncated SVD projection violates at these crossing points. Addressing this likely requires either a smoothed approximation to the hard rank-r truncation or an analysis that handles the discontinuity directly.

Adaptive rank selection. Developing a principled criterion for selecting consolidation rank—analogous to choosing the number of principal components via explained variance ratio—is an important open problem.

Token-space versus embedding-space perturbation balance. Context transplantation (token-space, discrete) and structured noise (embedding-space, continuous) probe different aspects of adapter reliability: cross-context generalization versus local representational robustness. Whether both are necessary, or whether one subsumes the other, requires empirical investigation. The prediction: context transplantation dominates for semantic-level consolidation (concept learning, relational structure), while embedding noise matters more for fine-grained representational stability (consistent behavior under small input variations).

Composite loss calibration. The contrastive coherence loss (§3.6) requires a temperature parameter τ that controls the sharpness of the contrastive signal, and the homeostatic decay rate η sets the timescale of forgetting—how quickly an unreinforced adapter relaxes toward exploration.

Both interact with the self-regulating ρ balance in ways that may require joint tuning. Additionally, the coherence floor λ_{base} prevents perpetual cramming under sustained moderate novelty, but the optimal value depends on how frequently the system encounters genuinely novel domains versus routine content.

Compute cost scaling. The consolidation loop (Algorithm 1) requires two forward passes per candidate context (one through the adapted model, one through the base model, for the anti-bias correction), plus one forward pass and one backward pass per chimeric dream sequence. Total per-cycle cost is therefore roughly $(2n + 2n')$ forward-pass equivalents, where n is the number of candidate contexts and n' is the number selected for dreaming. For toy experiments this is tractable. For production deployment with large models and frequent consolidation, this is the dominant cost bottleneck and will likely require approximations—e.g., amortizing the base-model forward passes across consolidation cycles, caching base-model losses for frequently-generated context types, or using a smaller distilled model as a proxy for the base-model loss computation.

Gradient materialization cost. The consolidation pipeline requires materializing the full gradient matrix $G_j \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ for each dream sequence, then computing a truncated SVD on the mean of these dense matrices. For the projection dimensions typical of large transformers, this dense materialization circumvents the memory efficiency that motivates using LoRA in the first place. Production-scale deployment will likely require either operating natively in the adapter's low-rank subspace (e.g., computing gradients with respect to the LoRA factors directly and projecting within that subspace) or employing randomized SVD algorithms that avoid materializing the full gradient matrix. The tradeoff between gradient-space fidelity and computational tractability is an important engineering question for any empirical implementation.

Rank growth under additive updates. Adding a rank-r SVD update to an existing rank-r adapter produces an adapter of rank up to $2r$. After k consolidation cycles, the adapter could in principle reach rank kr, violating the fixed-rank premise. Algorithm 1 addresses this via periodic rank re-projection: after each additive update, the accumulated adapters $B_Q A_Q$ and $B_V A_V$ are each re-decomposed via truncated SVD and refactored back into rank-r form, discarding the lowest-energy directions. This re-projection is itself an act of structured forgetting—the adapter retains only its r most energetic deformation directions, allowing less-used dimensions to be gradually overwritten. The computational cost of this re-projection (SVD of a $d_{\text{out}} \times d_{\text{in}}$ matrix each cycle) is non-negligible and may benefit from incremental or randomized approaches.

Anti-bias correction calibration and support limitations. The loss-ratio importance weight (Eq. 15) targets contexts where the adapter has made processing worse. However, a fundamental limitation applies: because candidate contexts are sampled from the adapted model, any region to which the adapted model assigns near-zero generative probability will never be proposed as a candidate, and no reweighting can recover it. The anti-bias correction successfully mines hard cases *within the adapted model's active support* but cannot audit genuinely avoided regions outside that support. A potential mitigation is to mix adapted-model samples with base-model samples during context generation, at the cost of introducing contexts that may be irrelevant to the adapter's actual deficiencies. The strength of the correction also requires modulation: too aggressive and the system spends all its dream-budget on pathological regions without consolidating routine improvements; too mild and confirmation bias persists. A temperature parameter on the loss ratio, possibly scheduled with adapter maturity, may be needed.

Initialization sensitivity and curriculum design. The self-reinforcing dynamics of the dreaming mechanism require existing adapter structure to function productively. At zero initialization, the entire structured-dreaming apparatus collapses to random consolidation, and the first few updates determine an initial receptive field that subsequent cycles reinforce. The Phase 1 curriculum (§5.7) addresses this by bootstrapping the adapter through externally supervised interactions, but the design of this curriculum—how diverse it must be, how many conversations it requires, and whether a shared initialization across users is preferable to per-user onboarding—is unresolved. The core risk is path-dependent snowballing: an unrepresentative initial receptive field that the self-reinforcing dynamics amplify rather than correct. Empirical investigation of how quickly the adapter's receptive field becomes representative enough for self-directed consolidation to be reliable is essential.

Privacy. A per-user LoRA adapter trained on conversational experience encodes experiential residue in a form that is not human-readable but may be extractable through adversarial probing. The relationship between adapter contents and original experiences requires careful analysis.

Base model migration. When the base model is retrained, existing per-user adapters may become misaligned. A mechanism for adapter migration across base model versions is needed.

Chimeric sequence construction. The paper specifies that salient content is "embedded" in alien contexts but does not commit to a method (insertion, replacement, interleaving). This design choice matters: insertion creates incoherent text whose gradients may reflect incoherence-handling rather than content processing; replacement alters positional encoding relationships; interleaving maintains local coherence but creates macro-level discontinuity. The quality of the consolidation signal depends critically on chimeric sequences being informative rather than confusing, and the optimal transplantation method is likely task-dependent. Empirical comparison of transplantation strategies is needed.

Adversarial and garbage input. The salience weighting (§5.2) flags high-prediction-error content as maximally consolidation-worthy. But the architecture has no mechanism to distinguish genuinely informative surprise from adversarial or garbage input—a user who deliberately feeds contradictory information would produce maximally salient content that the system would prioritize for consolidation. Similarly, the anti-bias correction (§5.3) cannot distinguish genuine blind spots from contexts that are appropriately outside the adapter's domain. Some form of input quality filtering, external to the consolidation loop, may be necessary for robust deployment.

Multi-layer considerations. The paper specifies adapters on Q and V projections but does not discuss which transformer layers should be adapted. Different layers encode different levels of abstraction; applying the same consolidation objective to all layers may be inappropriate. Lower layers may benefit from reconstruction-dominant consolidation (preserving fine-grained perceptual patterns) while higher layers may benefit from coherence-dominant consolidation (abstracting conceptual structure). Layer-specific scheduling, potentially with different ρ balances per layer, is an unexplored dimension of the architecture.

Maturity signal limitations. The architecture relies on a single scalar μ to drive salience weighting, noise anisotropy, loss balance, and phase transitions. This is a known architectural weakness. A narrow, large-norm deformation (strong commitments in few directions) and a broad, well-calibrated adapter (moderate commitments across many directions) can yield the same μ, despite representing very different developmental states—the former potentially pathological (over-commitment to one pattern), the latter healthy. The system would apply identical scheduling to both. Additionally, homeostatic decay during periods of routine inactivity erodes the deformation norm and drives μ down, potentially pushing a mature system back into exploratory behavior without corresponding evidence that its learned structure has become inadequate. The eigenvalue distribution of S_Q and S_V would distinguish these cases, but replacing μ with a spectral maturity signal would require redesigning the interpolation mechanisms in §5.2 and §5.5 and is deferred to future work.

Empirical validation. The framework is currently theoretical. Validation is needed on multiple fronts: whether structured dreaming outperforms random noise or direct replay; whether gradient-space SVD correctly identifies consolidation-worthy dimensions; whether the anti-bias correction successfully exposes adapter blind spots; whether the self-regulating loss balance produces the predicted reconstruction→coherence shift across consolidation cycles; whether the coherence loss produces genuinely more abstract and context-invariant representations than reconstruction alone; whether the shadow adapter dampens rotational dynamics in practice; and whether the system exhibits predicted behavioral properties (recognition of familiar content, reduced-dimensionality prediction error on repeated themes, graceful integration of novel experience without catastrophic forgetting).

9 Conclusion

We have proposed an architecture for transformer memory motivated by biological consolidation, specified in terms of existing transformer components, and analyzed using established mathematical frameworks. The central insight is that memory is not stored content but altered perception. The LoRA adapter does not record what happened; it changes how the system processes future experience in light of what happened. This is the computational analog of the biological insight that memory consolidation transforms episodic experience into structural modification of processing dynamics—you don't remember by retrieving stored records, you remember by being a system whose dynamics have been shaped by prior experience.

The key technical contributions are: (1) the formalization of the consolidation objective as gradient-space SVD with rank as an epistemic parameter; (2) a composite experiential loss with reconstruction and coherence components whose balance is self-regulated via spectral energy modulated by adapter maturity, motivated by the infant-sleep inversion and independently justified by cold-start dynamics, with homeostatic decay applied separately to the LoRA factors; (3) the specification of dream content generation as adapter-biased context transplantation with anti-bias correction, governed by a maturity-dependent schedule; (4) the stability analysis via constant-stepsize SA with bounded tracking as the target regime; (5) the derivation of four disambiguated safeguards from convergence requirements; and (6) a three-phase developmental trajectory derived from the architecture's own feedback dynamics.

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